



Analysis of GBA enzyme activity in plasma and CSF samples from PPMI

Ritesh Ravi, Jillian Kluss, Romeo Maciuca, and Jung Suh
Denali Therapeutics Inc

PPMI Project ID: 215
Version 2 – 10 June 2024

Summary

Plasma and CSF samples from the PPMI were analyzed for GBA enzyme activity in order to determine human GBA1 enzyme activity levels both in the blood and CSF of healthy control and Parkinson's disease (PD) patients which include pathogenic mutation carriers in *LRKK2* and/or *GBA*.

Revision 1. Version 1 contained the methodology of plasma sample processing and analysis. The current version includes the methodology of CSF sample processing.

Materials

Corning® 96-well Black Round Bottom Polypropylene Not Treated Microplate, without Lids, Nonsterile (Corning cat# 3356)
Microplate, 96 Well, PP, V-Bottom (Greiner Bio-One cat# 651201)
GBA Activity Buffer (100mM Phosphate Citrate Buffer (Sigma Aldrich cat# P4809), 0.5% Sodium Taurocholate (Sigma Aldrich cat# 86339), 0.25% Triton-X 100, pH =5)
GBA Stop Buffer (500mM Glycine, 300mM NaOH, pH 9.8)
4-Methylumbelliferyl β -D-glucopyranoside (Sigma-Aldrich cat# M36333)
Conduritol B Epoxide (prepare and store in 100mM aliquots) (Sigma Aldrich cat# C5424)

Method

Plasma sample preparation for GBA enzyme activity analysis

Transfer plasma samples into a 96 well V-bottom plate. Spin down plasma samples at 3500xg for 10 minutes at 4°C. Transfer plasma samples (leave about 20uL/sample to avoid precipitants) into a separate 96 well V-bottom plate.

Detecting GBA enzyme activity levels in human plasma

On the day of the assay add 1.4mL of 30mM 4-Methylumbelliferyl β -D-glucopyranoside (4-MU glucose) substrate to 38.6 mL of GBA activity buffer. In a separate V-bottom plate add control





human plasma sample (in quadruplicates). In one set of human plasma control samples add 2uL of Conduritol B epoxide (CBE). This will inhibit GBA1 activity and serve as a negative control for the assay. Next add 129uL/well of GBA activity buffer with 4-MU glucose substrate into the same V-bottom plate the controls are in and then add 4uL of prepared plasma samples (final concentration of 4-MU glucose substrate should be around 1mM). Mix all of the samples thoroughly and then transfer 100uL of the plasma in GBA activity buffer into a Corning® 96-well black round bottom Polypropylene plate. Seal and shake at RT for about 10 minutes. Place plate in a non-CO₂ 37°C incubator overnight.

On the next day perform a quick spin on the plate (1000xg for 60 seconds). Create a standard curve of 4- methylumbelliferone in GBA activity buffer and add to plate. Then add 100uL/well of GBA stop buffer. Read on plate reader using Ex 365/Em 445.

Detecting GBA enzyme activity levels in human CSF

On the day of the assay add 250uL of 500mM 4-Methylumbelliferyl β -D-glucopyranoside (4-MU glucose) substrate to 100 mL of GBA activity buffer. In a V-bottom plate add 100uL of GBA activity buffer with 4-MU substrate and add 25uL of CSF (final concentration of 4-MU glucose substrate is 1mM). Mix all samples thoroughly and transfer 100uL of CSF in GBA activity buffer into a Corning® 96-well black round bottom Polypropylene plate. A positive control was added to each plate. Seal and shake at RT for about 10 minutes. Place plate in a non-CO₂ 37°C incubator overnight. On the next day perform a quick spin on the plate (1000xg for 60 seconds). Create a standard curve of 4- methylumbelliferone in GBA activity buffer and add to plate. Then add 100uL/well of GBA stop buffer. Read on plate reader using Ex 365/Em 445.

Analyzing GBA enzyme activity levels in human plasma and CSF

4-Methylumbelliferyl β -D-glucopyranoside (4-MU glucose) is a synthetic GBA1 substrate which gets broken down to 4-Methylumbelliferone (4-MU) and glucose. 4-Methylumbelliferone can be read when excited at 365nm therefore the higher the read value of the 4-MU the more implied GBA activity in the samples.

Data analysis

The GCase enzyme activity assay for plasma was performed in eight different batches. A total of 3 pooled control and PD case plasma sample replicates were created and distributed evenly within and between batches to enable evaluation of inter-batch deviations. Within each batch, the pooled samples were analyzed six times each. Activity estimates derived were log transformed and values outside 2.5% to 97.5% of pooled samples activity distribution were excluded as outliers. Using these remaining values, a linear model was fitted with batch number as independent factor. Mean residuals derived from this linear model were used to correct for batch-dependent variances in the log transformed sample measurement data. Following batch correction, data were back transformed to linear scale and a 4-MU standard curve (Figure 1) was used to interpolate the A.U. values into concentration of 4-MU produced in each sample. To get a unit of substrate product produced over time in the sample, we divided the interpolated value



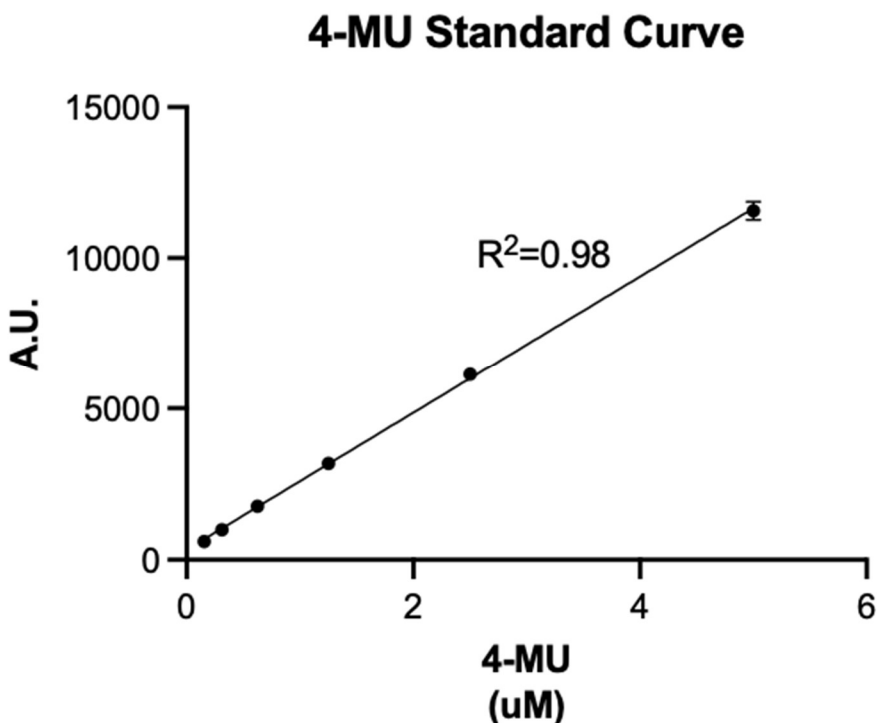


by the sample volume and incubation time. Using this methodology, we derived the final unit of “nM product/uL/hr”.

For the GCase enzyme activity assay in CSF, all samples were run on the same day. The samples were organized into 8 blocks and within each of the blocks 6 pooled sample QCs (quality control samples) were analyzed 6 times. The data was back transformed to a linear scale against a 4-MU standard curve to interpolate the A.U. into 4-MU produced in each sample. To get a unit of substrate product produced over time in the sample, we divided the interpolated value by the sample volume and incubation time. Using this methodology, we derived the final unit of “nM product/uL/hr”. No batch-to-batch correction was done for CSF samples.

Figure:

Figure 1: 4-MU standard curve



Data Availability

Analyte measurements from all study samples have been provided. Measured values in the 6 pooled quality control samples are available upon request.





References:

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